Temporally Stable Ultra-Compact Video Enhancement

Working manuscript: problem, protocol, and verified preliminary evidence

Internal research draft; authorship to be determined by the project team

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Abstract

Applying a compact image enhancer independently to video frames creates a quality–stability problem under a tight computation budget. We formulate the distinction between training-time temporal supervision and inference-time history, establish strong same-weight classical references, and document controlled evidence for adapting MobileIE. The completed experiments show that restoring the original expanded training parameterization improves a matched mixed-data baseline, while a fixed test-time local pooling conversion harms both evaluated domains. These findings constrain the next experiment; they do not establish a new video enhancement method. A factorial study of the root nonlinearity and a training-only temporal objective is currently running. This internal draft contains no final result for that candidate, independent final-test claim, or measured phone latency.

Draft status. This is a research manuscript under development, not a submission-ready paper. The final method, verified novelty, independent low-light video evaluation, and handset measurements remain incomplete. The numerical tables are generated from completed repository evidence. Training progress, partial checkpoints, and loss curves are excluded.

1 Problem and deployment constraints

An image enhancer maps a low-light observation x_t to an enhanced frame $z_t = f_\theta(x_t)$. A video system must also preserve legitimate scene changes, avoid noise-driven fluctuations, and fit its computation and memory budget. Low parameter count alone does not establish any of these properties. MobileIE [4] provides a useful compact starting point. Our 12-channel low-light deployment instance has 4,075 parameters; *4K parameters* does not mean a video resolution of 4K.

We distinguish a stateless predictor from a stateful one:

$$z_t = f_\theta(x_t), \qquad (z_t, h_t) = F_\theta(x_t, h_{t-1}).$$

A training objective may use neighbouring frames without introducing h_{t-1} at deployment. Conversely, a computation whose output changes for two histories with the same current frame cannot generally be represented exactly by a function of that current frame alone. Spatial convolution/BN folding does not eliminate this information dependency.

Our evaluation therefore reports quality, temporal error, parameters, convolution operations, online history, and measured pipeline latency as separate quantities. A small parameter change can still introduce full-resolution arithmetic or memory traffic. Inference motion estimation, sampling, padding, and buffering belong to the deployed pipeline even when they have no learned parameters. Server CPU measurements are not phone measurements.

1.1 The original nonlinear pathway

The retained MobileIE backbone consists of a convolutional head, a second feature block, channel/spatial gates, and an RGB readout [4]. Its feature self-transform (FST) has the form

$$q(u) = (w_1 u)(w_2 u) + b = au^2 + b,$$

where w_1, w_2 are shared scalars and b is channel-wise. Channel attention uses global average pooling followed by a 12×12 pointwise map and a sigmoid. The spatial gate depends on the channel-weighted feature maximum. The learned enhancement mapping, global statistics, and local nonlinear response can consequently interact.

For a scalar perturbation ϵ, the identity

$$q(u + \epsilon) - q(u) = 2au\epsilon + a\epsilon^2$$

motivates examining noise sensitivity at the root operator. If the perturbation has conditional mean zero, its conditional mean shift is $a\,\mathbb{E}[\epsilon^2]$. This elementary identity is not a new contribution. It neither proves that FST is the sole cause of video flicker nor establishes an unbiased correction for features after nonlinear convolutions, ISP processing, clipping, and quantization.

1.2 Known mechanisms and the remaining research question

Training-time reparameterization, channel multiplication, temporal regularization, and classical motion-compensated fusion are established tools. FastDVDnet [3], for example, learns multiframe denoising without an explicit flow estimator. TLC [1] addresses a different issue: the spatial-statistic mismatch between patch training and whole-image inference. Learnable safe rational activations also have clear precedents [2].

The unresolved question is whether an independently justified video adaptation can improve the quality–stability–cost trade-off over strong matched controls in this extremely compact setting. Neither assembling these known components nor obtaining a nonzero historical effect answers that question. The current candidate remains an experiment until its controlled results, external validity, and deployment costs support a defensible claim.

2 Candidate and controlled questions

The following candidate is being tested in the registered R44 study. Its performance and independent contribution have not been established.

2.1 A constrained root nonlinearity

We investigate the restricted rational response

$$q_\kappa(u) = \frac{au^2}{1 + \kappa(u/s)^2} + b, \qquad s > 0, \quad \kappa \geq 0.$$

Each of the two FSTs receives one additional scalar κ. This is a constrained member of the known rational-activation family [2]; we do not claim the rational form, safe denominator, or learnable activation as a new invention. At $\kappa = 0$, the original response is recovered. For positive κ,

$$q'_\kappa(u) = \frac{2au}{[1 + \kappa(u/s)^2]^2}, \qquad \sup_u |q'_\kappa(u)| = \frac{9|a|s}{8\sqrt{3\kappa}}.$$

The maximum occurs at $|u| = s/\sqrt{3\kappa}$. This scalar derivative bound does not establish a measured whole-network robustness or flicker bound; it also diverges when κ approaches zero.

The fixed scale s is the pre-FST feature RMS from eight predetermined training windows, computed separately for each layer and training seed. Calibration uses only training inputs and discards the temporary BN states. It fixes optimization units rather than selecting a scale on development quality. The learned denominator remains in deployment. The resulting predictor has 4,077 parameters, no temporal state in its raw path, and extra pointwise arithmetic compared with the 4,075-parameter square predictor. Only the spatial MBR convolutions and fixed-statistics BN are folded.

2.2 Training-only target-change compensation

Let W_t gather the previous frame into current-frame coordinates and let m_t be its validity mask. The candidate temporal objective is

$$\mathcal{L}_{\text{time}} = \frac{\sum_{t,p} m_t(p) \, \|[z_t - W_t z_{t-1}](p) - [y_t - W_t y_{t-1}](p)\|_2^2}{3 \sum_{t,p} m_t(p)}.$$

It is defined as zero when no points are valid. The frame reconstruction term still updates in that case. Subtracting the target change allows legitimate target motion or brightness change rather than assuming that the desired video is constant. Correspondences are estimated from training targets with the fixed half-resolution forward–backward estimator; they are not motion truth. Both predicted frames receive gradients. Deployment does not call this training correspondence estimator.

2.3 Factorial control

The fixed design crosses original square versus constrained rational FST with frame MSE versus $\mathcal{L}_{\text{MSE}} + 0.25\mathcal{L}_{\text{time}}$. All four models start from the same completed expanded-training checkpoint for each seed, with the same BN statistics. Adam is restarted for every arm. Each final model inherits 4,000 updates and receives 4,000 additional updates on the same prescribed windows. This controls for extra training, the objective, and the structural change; equal image/update budgets do not imply equal training FLOPs.

The primary structural screen compares the rational temporal model with the square model under the same temporal objective. A separate stateless deployment screen compares its raw output with both square controls' recursive postprocessing. A third screen compares all models with recursive postprocessing. A positive third screen cannot be reported as a stateless or flow-free result. Each registered screen includes both domains, both seeds, quality preservation, a temporal-error improvement requirement, and a worst-condition quality bound.

3 Experimental protocol and comparison scope

3.1 Development data

The current synthetic training pool contains 57 DAVIS training videos, excluding the three development sources *bear*, *lucia*, and *walking*. The synthetic development pilot uses 64 frames per source, at 480-pixel height, with two independently sampled noise conditions. There are three source videos and six conditions, not six independent videos. For a target RGB value y in $[0, 1]$, the degradation proxy is

$$v = 0.35y^{2.2}, \qquad x = Q_8(\text{clip}\,[\text{Poisson}(Pv)/P + \eta, 0, 1]), \quad \eta \sim \mathcal{N}(0, 0.008^2),$$

with $P \in \{20, 80\}$. Here Q_8 denotes conversion to the eight-bit observation. The present pilot adds no extra JPEG encoding, although the source frames are JPEG images. This is a controlled synthetic proxy, not a calibrated model of arbitrary RAW sensors or phone ISP video.

The fixed mixed-data study replaces half of each training batch with native crops from 12 verified paired ND-filter video sources in the project's OLiVES snapshot. Its four already-used paired-real development sources are videos 15, 47, 91, and 121, each evaluated on 16 frames with a 288×512 crop. Training and development source IDs are separated. Consecutive registered frame indices are available; independent sensor timestamps have not been verified. These daytime ND pairs must not be presented as natural nighttime validation.

Three original phone clips have additionally been inspected without synthetic degradation. They have no synchronized reference, so they support behavioural examples rather than reference PSNR or a perceptual flicker ranking. Independent final-test evaluation remains outstanding.

3.2 Quality and temporal measurements

Predictions are evaluated as floating-point RGB values after clipping to $[0, 255]$, before display rounding or video encoding. Sequence PSNR is the mean of full-frame RGB frame PSNR values,

$$\mathrm{PSNR}_{\mathrm{seq}} = \frac{1}{T} \sum_t 10 \log_{10} \frac{255^2}{\mathrm{MSE}_{\mathrm{RGB}}(z_t, y_t)}.$$

RGB SSIM uses an 11×11 Gaussian window with standard deviation 1.5 and the valid region obtained by excluding five boundary pixels. Edge MAE is the full-frame mean absolute error of horizontal and vertical luminance Sobel gradients, with the fixed gradient normalization used by the evaluation implementation.

For a fixed estimated reference flow and validity mask, temporal error is

$$E_{\mathrm{temp}} = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \frac{\sum_{p,c} m_t^{\mathrm{ref}}(p) \left| [z_t - W_t^{\mathrm{ref}} z_{t-1}]_c(p) - [y_t - W_t^{\mathrm{ref}} y_{t-1}]_c(p) \right|}{3 \sum_p m_t^{\mathrm{ref}}(p)}.$$

Here $\mathcal{T}$ contains frame pairs with nonempty valid masks. The sequence summary first averages each valid frame pair, then averages the available pairs. The reference mask excludes an eight-pixel boundary. Reference flow is estimated from targets and held fixed across compared methods; it is not ground-truth motion. This metric is not a sufficient perceptual definition of flicker. Evaluation uses fixed reference fields and absolute RGB error. Training uses half-scale estimated fields, differentiable continuous bilinear gathering, and squared error. The two are not numerically identical.

Domains and training seeds are reported separately before aggregation. The present two-seed pilot does not establish statistical significance. We retain per-condition failures alongside mean improvements.

3.3 Strong same-weight references

For each spatial anchor, let r_t denote the raw prediction. The finite reference uses the previous raw output, while recursive fusion uses the previous fused output:

$$z_t^{\mathrm{fin}} = \tfrac{1}{2} r_t + \tfrac{1}{2} W_t r_{t-1}, \qquad z_t^{\mathrm{rec}} = \tfrac{1}{2} r_t + \tfrac{1}{2} W_t z_{t-1}^{\mathrm{rec}}.$$

The formulas apply only on accepted correspondences; invalid points and the first frame use the current raw output. Both estimate motion from their own adjacent raw predictions with the fixed

Table 1: Completed anchor evidence: two-seed means on custom synthetic and paired ND development data. Historical rows differ in training; only the two R41 raw forms are a matched FP32 parameterization comparison.

Anchor/output	SYN PSNR	SYN E_{temp}	ND PSNR	ND E_{temp}
Dynamic synthetic (R34)	26.234	10.505	23.453	3.166
Fixed mixed Slim (R40)	25.134	11.363	24.861	3.628
Matched Slim FP32 (R41)	25.131	11.362	24.861	3.627
Expanded FP32 (R41)	26.186	10.007	25.847	3.630
Expanded + finite	26.394	7.286	25.734	2.804
Expanded + recursive	26.503	6.520	25.758	2.553
Earlier native training (R26)	–	–	27.138	3.313

half-resolution forward–backward estimator. Recursive fusion therefore also keeps the previous raw output. Neither reference requires a second enhancement CNN or a GT reset.

Published FastDVDnet weights [3] have been evaluated as fixed denoiser transfer, not as a retrained low-light video method. Its five-frame window uses two future frames. The documented pretraining source includes the three current DAVIS development IDs, so that transfer cannot establish unseen generalization. Same-domain retraining and broader published LLVE comparisons remain necessary.

3.4 Efficiency

Convolution MACs, pointwise work, history storage, and full pipeline time are distinct measurements. The original predictor has approximately 7.938 convolution GMAC at 1080p. This count omits non-convolution work. Online history counts use actual tensor storage rather than only logical channel slices. CPU timing is measured with two Torch threads and one OpenCV thread, and its workload and reference weights are stated explicitly. No handset latency is inferred from these measurements.

4 Verified preliminary evidence

The following results come from completed development experiments. They constrain method design; they are not evidence that the pending candidate has succeeded. The source manifest and generated tables accompany this draft.

4.1 A stronger training baseline

Table 1 separates historical anchors from the matched training-form comparison. In the matched FP32 experiment, restoring MBR/BN training improves synthetic raw PSNR over direct Slim fine-tuning by 0.968 and 1.142 dB for the two seeds. The corresponding paired-real gains are 0.171 and 1.799 dB. Both deploy to the same 4,075-parameter spatial class. Training parameterization is therefore a material baseline choice. This is known spatial reparameterization, not a new video mechanism.

The expanded control uses 49,978 training parameters, including an inactive frozen warm head; 47,239 parameters are optimized. It is a controlled fine-tuning study, not a complete reproduction of the official warm-up, loss, and long-schedule training. To preserve the pretrained numerical starting function, the wide raw convolution executes its preserved and additional output blocks

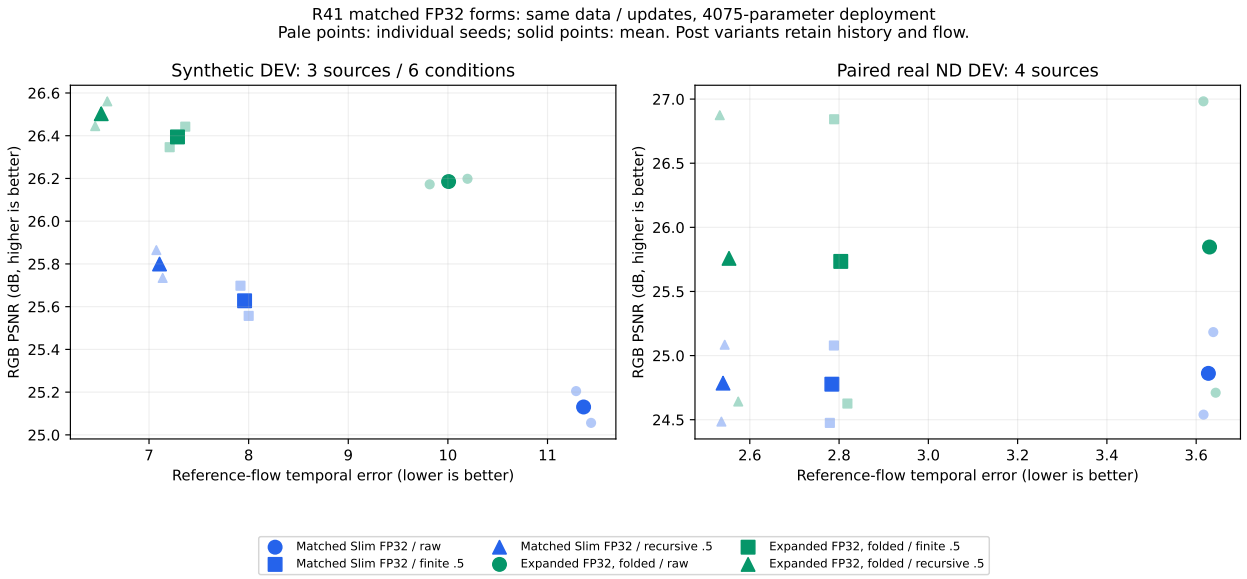


Figure 1: Completed matched FP32 training-form comparison. Pale points retain both individual seeds; solid points are their means. The real-domain spread and the quality–temporal trade-off are visible. Postprocessed outputs use motion and history; they are not stateless predictions.

Table 2: Classical references on the same R34 weights. Quality uses the full synthetic pilot. Resident CPU timing uses only the first 16 bear/P20 frames at 480×854, two Torch threads and one OpenCV thread. Ranges are run-average extrema, not confidence intervals or phone timings.

Output	PSNR	E_{temp}	History MiB	CPU ms/frame [range]
Raw	26.234	10.505	0.000	100.9 [80.9, 106.5]
Finite .5	26.466	7.543	4.691	184.1 [168.1, 188.5]
Recursive .5	26.610	6.735	9.382	186.5 [179.6, 192.5]

separately. This retains its parameters and mathematical operator but differs from the upstream single wide-kernel call.

The domain trade-off remains important. Expanded training approximately recovers the synthetic quality of the dynamic-synthetic anchor while improving its paired-real mapping. It still trails the earlier native training reference in paired-real PSNR. Those historical training schemes differ in several respects and do not isolate one causal factor. No single online model is obtained by taking the best weight separately for each domain.

Mean improvement does not imply reliable per-clip improvement. On paired-real video 121, expanded seed 0 loses 2.606 dB against its matched Slim control and shows sky colour bias in the fixed-frame inspection. Its four-source real mean is 24.710 dB, versus 26.983 dB for expanded seed 1. Raw paired-real temporal error is not consistently reduced. All source/seed pairs and all four metrics remain part of the record.

4.2 Classical recursion is a necessary reference

On the same strong dynamic-synthetic anchor, recursive fusion improves the finite reference's quality and temporal error (Table 2). Its persistent fused output is a meaningful information advantage, not a negligible implementation detail. The recursive reference cannot be replaced by a weaker

Table 3: Fixed known TLC192 versus the same R41 expanded weights. No retraining or window search.

Domain	Seed	Raw ΔPSNR	Raw temporal ratio
Synthetic	0	-0.305	1.0114
Synthetic	1	-0.415	1.0082
Paired ND	0	-2.312	1.0416
Paired ND	1	-2.343	1.0268

two-frame average when judging a proposed temporal model.

The measured CPU ranges overlap, so the small difference between the finite and recursive medians is not established as a significant speed difference. These timings concern the stated R34 weights and workload; they are not measurements of the later expanded anchor or the pending rational model. The extra stored RGB history and motion work remain part of deployment.

4.3 Fixed local statistics do not rescue this anchor

The fixed TLC [1] control replaces only the channel-pooling node with exact local averaging at a window of 192 pixels. Both development domains and both seeds lose raw PSNR; the paired-real temporal ratios also exceed the registered 1.02 upper bound (Table 3). Across the 20 raw seed–condition pairs, five PSNR deltas are positive, but all eight paired-real deltas are negative. The worst paired-real condition loses 3.716 dB.

The fixed instance and its local window search are stopped. This result does not prove that all local aggregation or global-statistic adaptations are ineffective. It also does not justify a speed claim: localizing the channel map adds approximately 0.299 convolution GMAC at 1080p, with prefix sums and memory traffic additional to that count.

4.4 Pending candidate results

The R44 factorial is still running at this manuscript snapshot. Its raw, finite, recursive, per-condition, and per-seed final results must be inserted only after both complete budgets and deployment-export gates finish normally. No current loss curve or intermediate checkpoint supplies this missing evidence.

5 Limitations and requirements for a complete paper

The present development pilot is small and repeatedly used. Synthetic noise, three DAVIS sources, and four paired daytime ND crops cannot establish general low-light video performance. The inspected phone clips show clipping and colour failures without synchronized ground truth. They do not replace a reference-based nighttime benchmark or a perceptual temporal study. Two training seeds expose variability but do not resolve it.

Motion-compensated target-residual error helps separate desired changes from unwanted differences, but its estimated reference flow can fail and its valid mask excludes difficult regions. Spatial quality, per-condition failures, scene-change behaviour, and visual examples must accompany it. Lower temporal error alone is insufficient: an inappropriate enhancement mapping may also look numerically stable.

The current published-weight tests are transfers with input-domain and pretraining-overlap limitations. A conference comparison still requires appropriate published temporal methods under a

documented common setting. The fixed classical references are strong controls, not full reproductions of all established video denoisers.

No candidate in this snapshot has a verified independent contribution or measured handset trade-off. A complete paper must establish the final method, explain its difference from rational activations and known temporal regularization, pass controlled structural/objective comparisons, measure the complete inference pipeline and state on a phone, and evaluate frozen decisions on genuinely independent data. If the registered candidate fails, its negative evidence should guide a new hypothesis rather than be rewritten as success. These requirements preserve the original research objective; the present manuscript is only the reusable foundation.

References

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